IMPROVED KISSING NUMBERS IN NINETEEN AND TWENTY-ONE DIMENSIONS

Abstract. We prove that the kissing numbers in 19 and 21 dimensions are at least 12276 and 31272. The best previously published lower bounds were 11948, due to Ho, and 29768, due to Cohn and Li. Both constructions start from the odd-sign kissing configurations of Cohn and Li and add weighted sign vectors. In 19 dimensions we cut the 20-dimensional configuration by a hyperplane, delete 184 of the points in it, and add 1792 points, the orbits of 14 integer vectors under a group of 128 sign changes; each is the projection of a 20-dimensional sign vector with reweighted head coordinates and a rescaled three-dimensional tail. In 21 dimensions we keep all 27720 points of Cohn and Li and add 3552 weighted sign vectors, the orbits of 222 integer vectors under a group of 16 sign changes, whose sign patterns come from cosets of both parities of a $[21, 9, 6]$ subcode of the punctured Golay code. Both proofs are exact computations, reduced by the symmetry, and an independent exact program checks all pairs of both configurations.

1. Introduction

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [4, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set X of unit vectors in $\mathbb{R}^n$ with $\langle x, y\rangle \le 1/2$ for all distinct $x, y \in X$. We write τ_n for this maximum, as in [4, Ch. 13]. It is known only for $n \in \{1, 2, 3, 4, 8, 24\}$; see [3, Table 1.1] for references.

For $17 \le n \le 23$ the best constructions were Leech's cross sections of the Leech lattice minimal vectors [6] until Cohn and Li [3] improved dimensions 17 through 21. In dimensions 19 to 21 their construction is as follows. Take the classical configuration built from a constant-weight code [7], and put an odd number of minus signs on each codeword instead of an even number. Every sign vector of a code orthogonal to the constant-weight code then has inner product at most 6 with the old points, while the kissing condition allows $\frac{1}{2}\sqrt{8n}$ (the sign vector has squared norm n and the old points 8), which is larger for $n \ge 19$. So these sign vectors can be added, one per word of a code with minimum distance at least $n/4$. Cohn and Li obtained $\tau_{19} \ge 11692$ and $\tau_{21} \ge 29768$ in this way, and Ho [5] found a larger code of 1280 words for dimension 19, reaching $\tau_{19} \ge 11948$.

Theorem 1.1. *We have $\tau_{19} \ge 12276$ and $\tau_{21} \ge 31272$.*

Table 1 compares these values with the best published lower bounds and the known upper bounds. The public construction of griffinwork40 [1] gives 31259 points in dimension 21. Our construction restores its 13 deleted base points, giving a gain of 13 over that repository result.

The mechanism is the slack between 6 and $\frac{1}{2}\sqrt{8n}$. Cohn and Li's added points are flat: each is a sign vector scaled to squared norm 8, and none of them touches an old point. We use this slack by changing the shape of the added points.

- In 19 dimensions (Section 3) the added points come from 20 dimensions. We cut the 20-dimensional configuration by the hyperplane orthogonal to the indicator vector of a set T of four coordinates. After an isometry, a point of that hyperplane has 16 *head*

2020 *Mathematics Subject Classification.* 52C17; 05B40; 94B05.

Key words and phrases. kissing number, spherical codes, Golay code.

TABLE 1. Kissing number bounds in dimensions 19 and 21, as of October 2026 [2]. The previous lower bounds are the best published ones. The upper bounds are the floors of numerical semidefinite programming values [8] and are not interval-certified.

dimension	previous lower bound	new lower bound	upper bound
19	11948 [5]	**12276**	24417 [8]
21	29768 [3]	**31272**	53524 [8]

coordinates, those outside T, and a three-dimensional *tail*. Projecting a sign vector into the hyperplane gives a *flat* point: head entries ± 1 and a short tail. Our added points keep the head signs of a sign vector but reweight the head entries and rescale the tail according to its direction. This lets two added points differ in only four head coordinates when their tails differ, which a code of minimum distance at least 5, such as Ho's, must avoid. We add 1792 points where Ho adds 1280. The long tails come too close to 192 old points, and we delete 184 of them; the other eight form one orbit of a group of 128 sign changes that maps the whole configuration to itself. The net gain over Ho is $1792 - 184 - 1280 = 328$.

- In 21 dimensions (Section 4) each added point carries its own positive weight on each coordinate. That freedom allows sign patterns from the even cosets of a $[21, 9, 6]$ code as well as from the odd ones, where an odd-weight choice of 2048 words uses only the four odd cosets. We use 3552 of the 4096 words of the punctured Golay code, against Cohn and Li's 2048, and delete nothing. The gain is $3552 - 2048 = 1504$.

The weights were found by numerical optimization and rounding. The 19-dimensional table was found by an AI agent outside this project, starting from added points of the shape described in Section 3.2, and was given in other coordinates; we state it in ours. Section 4.3 describes how the 21-dimensional weights were obtained. In both dimensions the proof uses only the final integer vectors.

The proofs depend on computer calculation in an essential way, unlike the proofs in [3]. Our added points are integer vectors with entries up to 4616 in 19 dimensions and between $1.49 \cdot 10^8$ and $2.48 \cdot 10^8$ in absolute value in 21 dimensions, and the theorems are statements about these explicit tables. The 14 representatives in dimension 19 are printed in Appendix A. In dimension 21 the 222 rows of 21 integers are too long to print; Appendix A gives their sign words, which determine the sign pattern of every added point, and four of the rows, and the full table accompanies the paper. Using their symmetry, the proofs reduce to lists of about 160 000 and 6.5 million exact integer inequalities, which we check by computer. A second program, written independently and without using the symmetry, checks all 75 343 950 and 488 953 356 pairs (Section 5).

2. The odd-sign configurations

We identify a binary word with its support. For $c \in \mathbb{F}_2^n$ we write $\mathrm{wt}(c)$ for its weight, $(-1)^c \in \{\pm 1\}^n$ for its sign vector, and $\lambda \circ q$ for the coordinatewise product of two vectors. We write $\mathbf{1}$ for the all-ones vector and $\mathbf{1}_T$ for the indicator vector of a set T of coordinates.

Let $\mathcal{O}$ be a set of words of length n with weight 8 and pairwise intersections of size at most 4. The *odd configuration* $B(\mathcal{O})$ consists of the $4\binom{n}{2}$ roots $\pm 2e_i \pm 2e_j$ and, for each $o \in \mathcal{O}$, the 128 vectors with entries ± 1 on o, zeros elsewhere and an odd number of minus signs [3, §2].

All have squared norm 8. We call these 128 vectors the *octad vectors* on o, and their set the *fiber* over o.

Lemma 2.1 (Cohn and Li)**.** *Distinct $b, b' \in B(\mathcal{O})$ satisfy $\langle b, b' \rangle \le 4$.*

Proof. Two distinct roots have inner product in $\{0, \pm 4, -8\}$. A root has two nonzero coordinates, each contributing at most 2 against an octad vector, so that inner product is at most 4. Two vectors on the same octad have odd sign patterns, so they differ in an even and positive number $d \ge 2$ of places and have inner product $8 - 2d \le 4$. Two vectors on different octads share at most four coordinates. □

The next lemma is the reason for odd signs. It is the weighted form of the argument in [3, §2].

Lemma 2.2. *Let $c \in \mathbb{F}_2^n$ meet every word of $\mathcal{O}$ in an even number of coordinates, let $q = (-1)^c$, and let $\lambda \in \mathbb{R}^n_{>0}$. Let b be an octad vector on $o \in \mathcal{O}$. Then q and b disagree in an odd number of coordinates of o, and*

$$\langle \lambda \circ q, b \rangle \;\le\; \sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j,$$

with equality for some b on o. In particular, for $\lambda = \mathbf{1}$ we have $\langle q, b \rangle = 6$ if q and b disagree in exactly one coordinate and $\langle q, b \rangle \le 2$ otherwise. For a root $b = \pm 2e_i \pm 2e_j$ the inner product is at most $2(\lambda_i + \lambda_j)$.

Proof. The sign vector q has an even number of minus signs on o, and b has an odd number. So they disagree in an odd number of coordinates of o, in particular in at least one, and the bound follows. Equality holds for the odd pattern that agrees with q except at one coordinate where λ is smallest. For $\lambda = \mathbf{1}$, d disagreements give $\langle q, b \rangle = 8 - 2d$ with d odd. □

Throughout we test compatibility in integers. For nonzero $x, y \in \mathbb{Z}^n$ the angle between them is at least 60° if and only if

$$\langle x, y \rangle \le 0 \quad \text{or} \quad 4\langle x, y \rangle^2 \le |x|^2 |y|^2. \tag{1}$$

We call such a pair *compatible*; the same test applies to rational vectors after scaling. If every pair from a finite set of nonzero rays is compatible, then the rays normalized to unit length form a kissing configuration, and distinct rays give distinct points because (1) excludes cosine 1. With $\lambda = \mathbf{1}$, Lemma 2.2 shows that the sign vector of a word that meets every word of $\mathcal{O}$ evenly is compatible with $B(\mathcal{O})$ when $6 \le \frac{1}{2}\sqrt{8n}$, that is, for $n \ge 18$, with contacts when $n = 18$.

3. Dimension 19

3.1. **The codes.** Coordinates are $1, \dots, 20$. Let $E \subseteq \mathbb{F}_2^{20}$ be the span of the twelve words

$$\{1,8,9,12,16,17,18,19\},\ \{2,10,11,14,15,17,18,20\},\ \{3,7,9,13,15,16,17,19\},$$
$$\{4,7,8,10,12,15,16,19\},\ \{5,10,12,13,15,16,17,18\},\ \{6,7,8,9,10,13,16,18\},$$
$$\{1,4,7,9\},\ \{1,5,6,18,20\},\ \{1,3,12,15,20\},\ \{1,10,13,19,20\},$$
$$\{1,3,5,6,7,13,14,15,18,20\},\ \{2,4,6,7,8,13,14,16,17,18\}.$$

On the first 19 coordinates these are Ho's twelve generators [5, §2], in his order. Our coordinate 20 is Ho's coordinate 24, the last column of his extension matrix, so we have added it to the words where that column has a 1. Ho shows that the full extension to length 24 is the extended binary Golay code [5, Prop. 2]. So E is the Golay code punctured at Ho's coordinates 20 to 23, and its dual $E^\perp$ is the Golay code shortened at the same four coordinates [9]. The 130

words of weight 8 in $E^\perp$ are the *octads*. They form the shortened Steiner system $S(5,8,24)$ with $n=20$ of [3, Table 2.1], so $B(\text{octads})$ is Cohn and Li's 17400-point configuration in $\mathbb{R}^{20}$, which we call B_{20}.

The five octads of the Golay code that contain four given coordinates partition the other twenty coordinates into five tetrads, and these six tetrads form a *sextet* [4, Ch. 10]. Puncturing at the four given coordinates turns those five octads into five words of weight 4 in E. We call these five words the *tetrads*.

The following facts are finite computations on the 4096 words of E.

Lemma 3.1. (1) *The weight enumerator of E is*

$$1+5z^4+64z^5+240z^6+320z^7+250z^8+640z^9+1056z^{10}\\ +640z^{11}+250z^{12}+320z^{13}+240z^{14}+64z^{15}+5z^{16}+z^{20}.$$

The five words of weight 4 are pairwise disjoint and cover all 20 coordinates.

(2) *Let $T=\{1,4,7,9\}$, one of the five tetrads. Two distinct octads meet in at most 4 coordinates. The octads meet T in 0, 2 or 4 coordinates, for 30, 96 and 4 octads respectively.*

The intersections in (2) are even because $T\in E$ and the octads lie in $E^\perp$. We call the other 16 coordinates the *head* H, and for $c\in E$ we write c_T and c_H for its parts on T and H.

3.2. **The configuration.** Let

$$R=\begin{pmatrix}1&1&-1&-1\\1&-1&1&-1\\1&-1&-1&1\end{pmatrix},$$

acting on the coordinates $(1,4,7,9)$ of T. The rows of $R/2$ are an orthonormal basis of $\mathbf{1}_T^\perp\cap\mathbb{R}^T$, so $(R/2)^{\mathsf{T}}(R/2)=I-J/4$. The map $\varphi(v)=(v_H, Rv_T/2)\in\mathbb{R}^{16}\times\mathbb{R}^3$ therefore satisfies

$$\langle\varphi(a),\varphi(v)\rangle=\langle a,v\rangle-\tfrac14\Big(\sum_{j\in T}a_j\Big)\Big(\sum_{j\in T}v_j\Big), \tag{2}$$

and it is an isometry from the hyperplane $\mathbf{1}_T^\perp$ onto $\mathbb{R}^{19}$. We write f_1,f_2,f_3 for the standard basis of the tail $\mathbb{R}^3$.

Base. There are 10668 points of B_{20} orthogonal to $\mathbf{1}_T$; this is the size of Cohn and Li's 19-dimensional base [3, Table 2.1]. They are the 480 roots inside H, the 12 roots $\pm2(e_i-e_j)$ with $i,j\in T$, the 3840 octad vectors on the 30 octads disjoint from T, the 6144 octad vectors on the 96 octads meeting T twice whose two T-entries have opposite signs, and the 192 octad vectors on the four octads containing T with two plus and two minus signs on T. We keep all of them except 184 of the last 192. Let $U=\{3,6,8,19\}$, so that $T\cup U$ is one of the four octads containing T, and let P be the set of the eight vectors

$$p_y=y+e_1+e_4-e_7-e_9,$$

where y has entries ±1 on U, zeros elsewhere and an odd number of minus signs. The base consists of the 10476 points of the first four kinds and the eight points of P, 10484 points b in all, and they enter as $\varphi(b)$. The images of the points of P are $\varphi(p_y)=(y_H,2f_1)$; two of them have inner product 0 if they differ in all four places of U and 4 otherwise.

Sign group. For a word $g\in E$ that misses T, let S_g change the signs of the coordinates in g. It preserves $\mathbf{1}_T^\perp$, and $\varphi\circ S_g=S_g'\circ\varphi$, where S_g' changes the signs of the head coordinates in

g; we write S_g for both. Let $\Gamma \subseteq E$ be the span of the seven words

$$\{2,5,10,11,13,15\},\quad \{2,5,10,16,17,20\},\quad \{2,5,12,14,15,17\},\quad \{2,10,13,14,16,18\},$$
$$\{2,3,6,10,14,15\},\quad \{2,3,8,10,11,17\},\quad \{2,3,11,13,16,19\}.$$

It has 128 words: 0, 48 of weight 6, 30 of weight 8, 48 of weight 10, and H. Every word of Γ misses T. The words of E that miss T form a group of order 256, which contains the four tetrads other than T, and Γ is one of the 16 subgroups of index 2 in it that contain none of these tetrads. The sign changes S_g, $g \in \Gamma$, form a group of order 128.

Lemma 3.2. *Every S_g, $g \in \Gamma$, maps the 10484-point base onto itself. On the 48 octad vectors of each fiber over an octad containing T that have two plus and two minus signs on T, the group $\{S_g\}$ has six orbits of size 8: the sets of points of that fiber with a given sign pattern on T. In particular P is one orbit.*

Proof. A word $g \in E$ meets every octad evenly, so S_g maps roots to roots and each fiber onto itself. Since g misses T, S_g fixes the entries on T, so it maps the 10668 points of B_{20} orthogonal to $\mathbf{1}_T$ onto themselves, and it preserves each of the five kinds listed above. Let $T \cup u'$ be an octad containing T. Its octad vectors with a given pattern on T are the eight sign vectors on u' with a fixed parity of minus signs, and S_g acts on them through $g \cap u'$, which has even size. For each of the four tetrads u', the map $g \mapsto g \cap u'$ takes all eight even subsets of u' as values on Γ; this is a finite computation. So $\{S_g\}$ acts transitively on these eight points. The set P is the class of the pattern $(+,+,-,-)$ on $T \cup U$, so it is an orbit, and the 184 deleted points are a union of orbits. Hence the base is mapped onto itself. □

New points. Let $r_1, \dots, r_{14} \in \mathbb{Z}^{19}$ be the vectors of Table 4 in Appendix A. Their entries have absolute values at most 4616, the only zero entries are in the tail, and their squared norms lie between 63 978 448 and 64 002 793. The new points are the $14 \cdot 128 = 1792$ vectors $S_g(r_i)$, $g \in \Gamma$.

The new points are weighted projections of sign vectors. Let $c \in E$ with $\mathrm{wt}(c_T) \in \{1,2,3\}$ and $q = (-1)^c$. The projected tail $t_0 = Rq_T/2$ satisfies $|t_0|^2 = 4 - \frac{1}{4}(\sum_T q_j)^2$ by (2). When $\mathrm{wt}(c_T)$ is odd, $t_0 \in \{\pm 1\}^3$; we call c *diagonal*. When $\mathrm{wt}(c_T) = 2$, $t_0 = \pm 2 f_k$; we call c an *axis* word. The 512 words with $\mathrm{wt}(c_T) \in \{0,4\}$ project to a zero tail, so there is no tail to rescale, and the construction does not use them. Let W be the span of the five tetrads, a code of size 32. The 3584 words with $\mathrm{wt}(c_T) \in \{1,2,3\}$ fall into 112 cosets of W. Each coset $c + W$ is a 5-cube, whose vertices are its 32 words and whose five directions add one tetrad each. We keep the words c for which $c + c_{\min}$ is a sum of an even number of tetrads, where $c_{\min}$ is the smallest word of the coset read as the integer $\sum_{j \in c} 2^{j-1}$. This is one of the two bipartition classes of each cube, 1024 diagonal and 768 axis words, 1792 in all. Two kept words never differ by a single tetrad. Two words that differ by a tetrad other than T have equal T-patterns, so their points would have head distance 4 and equal tails, with cosine above $1/2$ whatever the tail length. Two words that differ by T have the same head and opposite tails; with head entries ± 1 and tail of squared length τ the cosine is $(16 - \tau)/(16 + \tau)$, above $1/2$ for the diagonal tails used here ($\tau < 2.3$) though not for the longest axis tails.

For each new point x there is a kept word c such that x has the sign of $q = (-1)^c$ on every head coordinate and, among the 14 tail directions $\{\pm 1\}^3 \cup \{\pm 2 f_k\}$, its tail is closest in angle to t_0. Every kept word occurs exactly once; this is a finite computation. So $r_1, \dots, r_6$ come from axis words and $r_7, \dots, r_{14}$ from diagonal words, and Γ permutes the kept words. For a

TABLE 2. The blocks of the 19-dimensional configuration. The base points are given by their preimages under φ.

block	size
roots in H	480
roots in T	12
octad vectors, octad disjoint from T	3840
octad vectors, octad meeting T twice	6144
the orbit P on the octad $T \cup U$	8
new points from diagonal words	1024
new points from axis words	768
total	12276

representative $r = (r_H, r_{\text{tail}})$ define the *tail factor*

$$\alpha(r) = \frac{|r_{\text{tail}}|}{|t_0|}\Big(\frac{1}{16}\sum_{j\in H} r_j^2\Big)^{-1/2},$$

the tail length relative to that of the flat point with the same head size. It lies between 0.834 and 0.872 for diagonal and between 1.235 and 1.414 for axis words. Remark 3.6 explains these ranges.

We do not have a closed form for these weights. They were found by numerical optimization with the symmetry under Γ imposed: a penalty on pairs with cosine above $1/2$ was minimized over all 19 coordinates of the 14 representatives, 266 real variables, and the result was normalized to length about 8000 and rounded to integers. The proof does not depend on how they were found.

Table 2 lists the blocks. The total is $10484 + 1792 = 12276$.

3.3. **Proof.**

Proposition 3.3. *The 10484 base points and the 1792 vectors $S_g(r_i)$ are 12276 pairwise compatible nonzero vectors spanning $\mathbb{R}^{19}$. Hence $\tau_{19} \geq 12276$.*

Each S_g is orthogonal and $S_g S_{g'} = S_{g+g'}$. With Lemma 3.2, every pair involving a new point is carried by some S_g to a pair in one of these finite lists:

(i) the $\binom{14}{2} \cdot 128 + 14 \cdot 127 = 13\,426$ products $\langle r_i, S_g r_j\rangle$ with $i \leq j$, excluding $i = j$, $g = 0$;
(ii) the $14 \cdot 10484 = 146\,776$ products $\langle r_i, b\rangle$ with b in the base.

Lemma 3.4. *Let $L = 50\,000$. For every product $\langle x, y\rangle$ in lists (i) and (ii), $\langle x, y\rangle \leq (\frac{1}{2} - \frac{1}{L})|x||y|$. In particular every such pair satisfies* (1).

Proof. This is a finite computation in exact integer arithmetic. For $\langle x, y\rangle > 0$ the inequality is equivalent to $(2L\langle x, y\rangle)^2 \leq (L-2)^2|x|^2|y|^2$, and we verified this for all 160 202 products. As an independent check, the program of Section 5, written separately and without the symmetry reduction, verified (1) for all 75 343 950 pairs of the configuration. □

Proof of Proposition 3.3. The base consists of points of B_{20} orthogonal to $\mathbf{1}_T$, so base pairs are compatible by Lemma 2.1 and (2). Pairs involving a new point are compatible by Lemma 3.2 and Lemma 3.4. The base contains all 492 roots, which span $\mathbb{R}^{19}$. □

This proves Theorem 1.1 for $n = 19$.

3.4. **Remarks.**

Remark 3.5. The smallest value of $\frac{1}{4}-\cos^2$ over pairs involving a new point is $5393/170666864 \approx 3.16\cdot10^{-5}$ for base–new pairs and about $2.17\cdot10^{-5}$ for new–new pairs, where the largest cosine is $31\,997\,485/63\,997\,753 \approx 0.4999783$, between r_3 and one of its images. No pair involving a new point has cosine within 10^{-5} of $1/2$; 1792 base–new and 3456 new–new pairs are within 10^{-4}. All 2 389 952 contacts are between base points, and 3368 of them involve a point of P.

Remark 3.6. Consider a flat new point $(q_H, \alpha t_0)$, with all head entries ±1 and tail factor α. A diagonal point against an octad vector on an octad disjoint from T needs $36 \le 2(16+3\alpha^2)$, so $\alpha \ge \sqrt{2/3} \approx 0.8165$. Two axis points with the same pattern on T at head distance 6 need $4+4\alpha^2 \le \frac{1}{2}(16+4\alpha^2)$, so $\alpha \le \sqrt{2} \approx 1.4142$. A diagonal and an axis point at distance 1 on T and 4 on H, with tail factors α and β and tails making an acute angle, have inner product $8+2\alpha\beta$ against the bound $\frac{1}{2}\sqrt{(16+3\alpha^2)(16+4\beta^2)}$. With $\alpha=\beta$ the inner product exceeds the bound for every α in $[\sqrt{2/3},\sqrt{2}]$ (at $\alpha=1$ it is 10 against 9.75), so the two factors cannot be equal. The table has the diagonal factor near 0.83 and the axis factor above 1.2.

Axis factors above 1 are what the deleted points cannot tolerate. A point b on an octad containing T with two signs of each kind on T, and a flat axis point with $q_T = b_T$ whose sign vector disagrees with b in exactly one place of the octad, have inner product $6+4(\alpha-1)$ by (2), above the bound $\sqrt{2(16+4\alpha^2)}$ for α near 1.4. For the points of P these axis points are the 128 with tail along $+f_1$, each conflicting with four points of P; they form the orbit of r_6. Since Γ moves any point of P to any other, keeping one point of P forces all eight, and the table must change in the same way in either case. In the table, r_6 has tail factor 1.235, and its entries on $3, 6, 8, 19$ are 1453 in absolute value against 1777 on the other head coordinates, a ratio of 0.82. The other axis representatives have tail factor 1.400 (tail along $-f_1$) or 1.414 (tail along $\pm f_2$, $\pm f_3$), and the diagonal ones 0.834 or 0.872 as the first tail coordinate is negative or positive. Eight of the 14 representatives have all 16 head entries equal in absolute value, and the head entries of each of the others except r_6 vary by at most 1.6%.

Remark 3.7. For each of the 184 deleted points, the largest cosine with a new point lies between 0.5286 and 0.5525. By Lemma 3.2, a point put back on any fiber forces its whole orbit of 8 under Γ, if the configuration is to stay invariant under Γ. We do not know whether a second orbit can be kept.

Remark 3.8. Among groups of sign changes S_g with $g \in E$ missing T, the group Γ is as large as possible for this table. These words g form a group of order 256, and it contains the four tetrads u' other than T. If $S_{u'}$ were in the group, a new point x and $S_{u'}x$ would both be in the configuration, with cosine $1-2m/|x|^2$, where m is the part of $|x|^2$ on u'; this is at most $1/2$ exactly when $m \ge |x|^2/4$. For every r_i and every such u', $m < 0.23|r_i|^2$, so the cosine is above 0.54, and the same holds for any table whose head entries are nearly equal in absolute value. A subgroup of the 256-word group that contains none of the four tetrads has index at least 2, so order at most 128, and with it the table needs at least $1792/128 = 14$ representatives.

The numbers in Remarks 3.5–3.8 were computed by computer; the fractions are exact and the decimals are rounded.

4. Dimension 21

4.1. **The codes.** Coordinates are $1,\dots,24$, and the construction lives on the first 21. Let $\mathcal{G}$ be the extended binary Golay code: the cyclic code of length 23 generated by $1+t+t^5+t^6+t^7+t^9+t^{11}$, with the coefficient of t^k at coordinate $k+1$, extended by an overall parity bit at

coordinate 24 [4, Ch. 3, §2.8]. Let $\mathcal{O}$ be the 210 octads of $\mathcal{G}$ that vanish on coordinates 22, 23 and 24. Then $B = B(\mathcal{O})$ is Cohn and Li's 27720-point configuration in $\mathbb{R}^{21}$.

Let D be $\mathcal{G}$ punctured to the first 21 coordinates, a $[21, 12, 5]$ code. Because $\mathcal{G}$ is self-dual and the octads in $\mathcal{O}$ vanish on the punctured coordinates, every word of D meets every octad of $\mathcal{O}$ evenly. We call the 21 words of weight 5 in D *lines*; any two lines meet in exactly one coordinate, and they form a projective plane of order 4 on the 21 coordinates. Two sign vectors of distinct words of D at distance d have cosine $1 - 2d/21$, which exceeds $1/2$ only for $d = 5$, where it is $11/21$. Cohn and Li add 2048 sign vectors, from a linear subcode of D with minimum distance 6 [3, §2]. The 2048 odd-weight words of D, a translate of the even-weight subcode, give another valid choice: the sum of two odd words is even, so no two of them are at distance 5.

Let $C \subseteq D$ be the span of the nine words (bit j of the binary expansion is coordinate $j+1$)

$$1559917,\ 15163,\ 774651,\ 1540761,\ 485602,\ 408336,\ 678478,\ 1748752,\ 1231514.$$

It is a $[21, 9, 6]$ code with only even weights, and D/C has eight cosets of 512 words. Four are even, with representatives 0, 1393, 2139, 3370, and contain no lines. Four are odd, with representatives 696, 1993, 2787, 3986, and contain 6, 3, 6 and 6 lines. These are finite computations on the 4096 words of D.

The following lemma is not used in the proof. It explains why the new points must be weighted.

Lemma 4.1. *Let c, c' be distinct words of D. If they lie in the same coset of C, or in two cosets of the same parity, then $\mathrm{wt}(c + c') \geq 6$. If $c + C$ and $c' + C$ have opposite parity, then their sum is an odd coset, which contains a line; so $\mathrm{wt}(c_1 + c_1') = 5$ for some $c_1 \in c + C$ and $c_1' \in c' + C$.*

Proof. In the first case $c + c'$ lies in C or in an even coset, and the even words of D have weight at least 6. In the second case $c + c'$ lies in an odd coset, and every odd coset contains a line. □

So flat sign vectors from an even coset and an odd coset conflict at cosine $11/21$.

4.2. **Weighted sign vectors.** The base limits how far a weight vector can depart from flat. Let ℓ be a line, and put $\lambda_j = \sqrt{2/5}$ for $j \in \ell$ and $\lambda_j = 1$ otherwise, so $|\lambda|^2 = 18$. For $c \in D$ the point $\lambda \circ (-1)^c$ has, by Lemma 2.2, inner product at most $\sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j$ with the octad vectors on o. An octad meets ℓ in 0, 2 or 4 coordinates, because $\ell \in D$, and the bound is then $8 - 2 = 6$, $6 + 2\sqrt{2/5} - 2\sqrt{2/5} = 6$, or $4 + 2\sqrt{2/5} < 6$. This is at most $\frac{1}{2}\sqrt{8}|\lambda| = 6$. So among weight vectors equal to 1 off a line and constant on it, $\sqrt{2/5}$ is the smallest weight on the line that the base allows. One weighted line does not remove the conflict of Lemma 4.1. If both points carry this λ and their words differ by a line $\ell' \neq \ell$, which meets ℓ in one coordinate, the cosine drops from $11/21 \approx 0.524$ to $23/45 \approx 0.511$; if they differ by ℓ itself it rises to $7/9$. The 222 weight vectors below act on many lines at once.

In general, let $c \in D$, $q = (-1)^c$ and $\lambda \in \mathbb{R}^{21}_{>0}$. Lemma 2.2 shows that $\lambda \circ q$ is compatible with every point of B exactly when

$$\sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j \leq \sqrt{2}\,|\lambda| \ \text{ for all } o \in \mathcal{O}, \qquad \lambda_i + \lambda_j \leq \tfrac{1}{\sqrt{2}}|\lambda| \ \text{ for all } i \neq j. \tag{3}$$

The hypothesis $c \in D$ is what makes the first condition exact. Two points $\lambda \circ q$ and $\mu \circ q'$ have inner product $\sum_j \lambda_j \mu_j (-1)^{(c+c')_j}$, which depends only on the weights and on $c + c'$.

4.3. **The configuration.** *Base.* All of B, the 27720 points of Cohn and Li.

Sign group. Let $\Sigma \subseteq D$ be the span of 92361, 20689, 346115 and $w = 14723$, with supports

$$\{1,4,7,8,12,14,15,17\}, \quad \{1,5,7,8,13,15\}, \quad \{1,2,12,15,17,19\}, \quad \{1,2,8,9,12,13,14\}.$$

For $h \in \Sigma$ let S_h change the signs of the coordinates in h. The group $\{S_h\}$ has order 16. The first three generators span a subgroup $\Sigma_0 \subseteq C$ of order 8; the word w has odd weight, so $w \notin C$. Adding w swaps the cosets of C in the pairs $0 \leftrightarrow 696$, $1393 \leftrightarrow 1993$, $2139 \leftrightarrow 2787$ and $3370 \leftrightarrow 3986$, each pair joining an even and an odd coset.

New points. Let $s_1, \dots, s_{222} \in \mathbb{Z}^{21}$ be the representatives in the accompanying data, one for each orbit of the new points under $\{S_h\}$, chosen with the smallest sign word as an integer and numbered in that order. They have no zero entries, their entries have absolute values between 149 081 678 and 247 947 863, and their squared norms are within $3.1 \cdot 10^{-9}$ of 10^{18} in relative terms. The new points are the $222 \cdot 16 = 3552$ vectors $S_h(s_i)$, $h \in \Sigma$. Appendix A lists the 222 sign words, which fix the sign pattern of every new point, and prints four of the representatives in full.

The sign word of a point x with no zero entries is the word c with $x_j < 0$ exactly for $j \in c$. The sign words of the new points are 3552 distinct words of D, and each orbit uses one coset of Σ in D. In terms of the cosets of C:

- every word of the cosets of 1393, 1993, 2139 and 2787 is the sign word of exactly one new point, 512 on each; these are 128 orbits, 64 for each of the two pairs;
- the cosets of 0, 696, 3370 and 3986 carry 376 new points each, from 47 orbits for each of the two pairs;
- the 34 cosets of Σ in D that carry no new point lie in these last four cosets of C, 136 words in each.

So both parities occur in every orbit, and Lemma 4.1 shows why the points must be weighted. After scaling to $|\lambda|^2 = 18$, the entries of each representative lie between 0.68 and 1.14 times their root mean square. We do not have a closed form for these weights. The starting weights and the 222 sign orbits come from griffinwork40's 31259-point construction [1], which reweights the 192 orbits of our earlier 30779-point configuration and adds 30 orbits. Seven of its 13 omitted base points fit directly. To restore the remaining six, the final weights were obtained by numerical optimization with the sign words fixed and the symmetry under Σ imposed, finishing with sequential linear programming over all 222 orbits, linearizing the conditions (3) and the pairwise conditions at the current weights, until every point of B fits; the result was normalized to length about 10^9 and rounded to integers. The proof does not depend on how the weights were found.

The total is $27720 + 3552 = 31272$.

4.4. **Proof.**

Lemma 4.2. *Every S_h, $h \in \Sigma$, maps B onto itself.*

Proof. Every word of Σ lies in D, so it meets every octad of $\mathcal{O}$ in an even number of coordinates. A sign change S_h maps roots to roots. On an octad vector with support o it changes an even number of signs, so the number of minus signs on o stays odd. So $S_h(B) = B$. □

Proposition 4.3. *The 27720 points of B and the 3552 vectors $S_h(s_i)$ are 31272 pairwise compatible nonzero vectors spanning $\mathbb{R}^{21}$. Hence $\tau_{21} \geq 31272$.*

Each S_h is orthogonal and $S_h S_{h'} = S_{h+h'}$. Hence

$$\langle S_h s_i, S_{h'} s_j \rangle = \langle s_i, S_{h+h'} s_j \rangle, \qquad \langle S_h s_i, b \rangle = \langle s_i, S_h b \rangle.$$

With Lemma 4.2, every pair involving a new point is determined by one of these finite lists:

(i) the $\binom{222}{2} \cdot 16 + 222 \cdot 15 = 395\,826$ products $\langle s_i, S_h s_j \rangle$ with $i \le j$, excluding $i = j$, $h = 0$;
(ii) the $222 \cdot 27720 = 6\,153\,840$ products $\langle s_i, b \rangle$ with $b \in B$.

Lemma 4.4. *Let $L = 250\,000$. For every product $\langle x, y \rangle$ in lists (i) and (ii), $\langle x, y \rangle \le (\frac{1}{2} - \frac{1}{L})|x||y|$. In particular every such pair satisfies* (1).

Proof. This is a finite computation. For $\langle x, y \rangle > 0$ the inequality is equivalent to $(2L\langle x, y \rangle)^2 \le (L-2)^2 |x|^2 |y|^2$, and we verified it for all 6 549 666 products in exact integer arithmetic. As an independent check, the program of Section 5, written separately and without the reduction, verified (1) for all 488 953 356 pairs of the configuration. □

Proof of Proposition 4.3. Base pairs are compatible by Lemma 2.1. Pairs involving a new point are compatible by Lemmas 4.2 and 4.4. The base contains all 840 roots, which span $\mathbb{R}^{21}$. □

This proves Theorem 1.1 for $n = 21$.

4.5. **Remarks.**

Remark 4.5. The smallest value of $\frac{1}{4} - \cos^2$ over pairs involving a new point is

$$\frac{2929231201163}{666666666060433304} \approx 4.39 \cdot 10^{-6}$$

for base–new pairs, attained at an octad vector, and about $4.04 \cdot 10^{-6}$ for new–new pairs, where the largest cosine is about 0.4999960. All 12 988 080 contacts are between base points; they are the contacts of B.

Remark 4.6. With the base held fixed, each new point may be moved independently to any unit vector within chord distance $1/(4L) = 10^{-6}$ of its position, and the result is again a kissing configuration: if x, y move to $x+\delta_1$, $y+\delta_2$, the inner product changes by $\langle \delta_1, y \rangle + \langle x, \delta_2 \rangle + \langle \delta_1, \delta_2 \rangle$, at most $1/(2L) + 1/(16L^2) < 1/L$. So with B fixed, the kissing configurations near this one contain an open subset of $(S^{20})^{3552}$, of dimension $3552 \cdot 20 = 71040$. The same argument applies in dimension 19 with $L = 50\,000$ and chord distance $5 \cdot 10^{-6}$. One point on every word of D would give $27720 + 4096 = 31816$ points. We do not know whether any of the 34 unused cosets of Σ can carry an orbit.

The numbers in Remarks 4.5 and 4.6 were computed by computer; the fractions are exact and the decimals are rounded.

5. Verification

What is proved by hand and what by computer. By hand: Lemmas 2.1 and 2.2, the symmetry reductions (Lemmas 3.2 and 4.2) and the choice of the lists (i) and (ii). By computer, in exact integers: Lemma 3.1, the coset and orbit counts of Sections 3 and 4, Lemmas 3.4 and 4.4, and the all-pairs check below.

All pairs. One program, written independently of the programs that built the configurations, checks each configuration on all pairs, with no symmetry reduction and no floating point in any decision. It confirms the number of points and that every point is nonzero. For every pair of distinct points x, y it decides (1), that is $\langle x, y \rangle \le \frac{1}{2}|x||y|$: the coordinates are integers, and when $\langle x, y \rangle > 0$ it compares $4\langle x, y \rangle^2$ with $|x|^2 |y|^2$ exactly, in 128-bit integers. It also confirms that no two points have the same direction. Table 3 gives the results. Every pair passes, and in each configuration the largest cosine is exactly $\frac{1}{2}$. This check alone proves Theorem 1.1. The proofs by hand explain why the configurations work.

TABLE 3. The exact all-pairs check.

dimension	points	pairs	result
19	12276	75 343 950	pass
21	31272	488 953 356	pass

6. Conclusion

We have shown that $\tau_{19} \geq 12276$ and $\tau_{21} \geq 31272$, improving the best published lower bounds of 11948 and 29768. In 19 dimensions two questions are open. Can a second orbit of eight deleted points be kept, under Γ or under a smaller group with a larger table? And does the table have a closed form? Remark 3.6 suggests one: outside the orbit of r_6 it is close to flat points with one tail factor for diagonal and one for axis words, up to about 5%. In 21 dimensions, can any of the 34 unused cosets of Σ in D carry an orbit? And does weighting sign vectors from cosets of both parities help in 20 or 22 dimensions?

Use of artificial intelligence

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions and draft this paper.

Data availability

The coordinates and an exact checking program accompany this paper.

Appendix A. Representatives

Table 4 lists the 14 representatives of Section 3.2, in the order of the accompanying data. The axis representatives $r_1, \ldots, r_6$ have tails along $-f_1$, $-f_2$, $+f_3$, $-f_3$, $+f_2$ and $+f_1$; r_6 is the representative of Remark 3.6. Each is written as two rows, the first with the ten head coordinates 2, 3, 5, 6, 8, 10 to 14 and the second with the six head coordinates 15 to 20 and the tail.

TABLE 4. The 14 representatives r_i in dimension 19. Columns $2, \ldots, 20$ are the head coordinates, and f_1, f_2, f_3 the tail.

	2	3	5	6	8	10	11	12	13	14
	15	16	17	18	19	20	f_1	f_2	f_3	
r_1, axis	1638	1638	−1638	−1638	−1638	1638	1638	−1638	1638	1638
	−1638	1638	−1638	1638	1638	1638	−4588	0	0	
r_2, axis	−1633	1633	1633	−1633	−1633	−1633	−1633	1633	1633	1633
	−1633	1633	1633	1633	1633	1633	−102	−4616	1	
r_3, axis	1633	1633	1633	−1633	−1633	−1633	1633	1635	1633	1633
	1633	1635	−1633	1633	1633	1633	−101	0	4616	
r_4, axis	−1633	−1633	−1633	−1633	−1633	−1633	−1633	−1633	1633	1633
	1633	1633	−1633	1633	−1633	1633	−100	0	−4616	
r_5, axis	1633	−1633	−1633	−1633	−1633	−1633	1633	−1633	1633	1633
	−1633	1633	1633	1633	−1633	1633	−102	4616	1	
r_6, axis	−1777	−1453	1777	−1453	−1453	1777	−1777	1777	1777	1777
	−1777	1777	−1777	1777	−1453	1777	4203	0	0	
r_7, diagonal	−1881	−1881	1881	−1881	−1881	1881	1881	−1881	1881	1881
	1881	1881	−1881	1881	1881	1881	−1568	−1568	−1568	
r_8, diagonal	1881	−1881	1881	−1881	−1881	1881	−1881	−1881	1881	1881
	−1881	1881	1881	1881	1881	1881	−1568	1568	1568	
r_9, diagonal	−1858	−1885	−1885	−1858	−1858	−1856	1885	1856	1885	1858
	−1885	1856	−1856	1885	1885	1885	1593	−1649	1649	
r_{10}, diagonal	1881	−1881	−1881	−1881	−1881	−1881	−1881	1881	1881	1881
	1881	1881	1881	1881	1881	1881	−1568	−1568	1568	
r_{11}, diagonal	−1885	1858	1884	−1885	−1885	−1857	1859	−1857	1884	1885
	1884	1857	1857	1884	−1858	1859	1592	1650	−1650	
r_{12}, diagonal	1881	1881	1881	−1881	−1881	−1881	−1881	−1881	1881	1881
	−1881	1881	−1881	1881	−1881	1881	−1568	1568	−1568	
r_{13}, diagonal	−1859	1859	−1884	−1884	−1884	1884	1885	1857	1884	1859
	−1857	1857	1884	1857	−1858	1885	1592	−1650	−1650	
r_{14}, diagonal	1885	1885	−1885	−1858	−1858	1885	−1858	1856	1885	1885
	1856	1856	−1885	1856	−1885	1858	1593	1649	1649	

For dimension 21, the sign words of $s_1, \ldots, s_{222}$, as integers with bit j at coordinate $j+1$ and in the order of Section 4.3, are

0, 696, 1393, 1993, 2139, 2787, 3370, 3986, 4279, 4623, 5574, 6014, 6380, 6740, 7581, 32973, 33397, 34236, 34564, 35374, 36327, 36703, 36986, 37570, 38155, 38835, 38945, 39577, 40272, 40936, 131470, 131894, 132679, 133589, 134308, 134684, 135481, 136065, 136944, 137570, 138259, 138923, 164163, 164914, 165514, 166168, 166816, 167017, 167633, 168436, 168780, 169093, 169533, 170415, 170775, 171622, 524786, 525130, 525443, 525883, 526761, 527121, 527968, 528709, 529460, 530060, 530718, 531366, 531567, 532183, 557375, 557959, 558158, 558838, 559460, 560092, 560149, 560813, 561544, 561968, 562753, 563667, 564386, 564762, 655484, 656068, 656653, 658079, 658774, 659438, 659659, 660083, 660922, 661648, 662056, 663009, 663385, 688305, 688649, 689600, 690040, 690410, 690770, 691611, 692926, 693623, 694223, 694365, 695013, 695596, 696212, 1048924, 1049572, 1049645, 1050261, 1050887, 1051766, 1052366, 1053163, 1053523, 1053850, 1054242, 1055152, 1055496, 1055937, 1056377, 1081745, 1082592, 1082968, 1083850, 1084274, 1084603, 1084931, 1085734, 1086551, 1087215, 1087869, 1088453, 1089204, 1180266, 1181091, 1181467, 1181833, 1182257, 1183224, 1183552, 1183845, 1184477, 1185044, 1185708, 1185854, 1186438, 1187151, 1212447, 1213095, 1213806, 1214422, 1215228, 1215797, 1216397, 1216680, 1217040, 1218009, 1218401, 1219147, 1219970, 1220410, 1573038, 1573398, 1574367, 1574759, 1575501, 1576324, 1576764, 1576985, 1577633, 1578344, 1579770, 1580339, 1580939, 1606363, 1606930, 1607594, 1607736, 1608320, 1609033, 1609940, 1610348, 1611173, 1611549, 1611919, 1612343, 1613310, 1613638, 1704224, 1704856, 1705041, 1705705, 1706363, 1706947, 1707698, 1708439, 1708847,

1709286, 1709662, 1710540, 1710964, 1711293, 1711621, 1737197, 1737557, 1737884, 1738276, 1739190, 1739975, 1740415, 1741146, 1741794, 1742483, 1743105, 1743984, 1744584.

Each is the smallest word of its coset of Σ; the sign words of the 3552 new points are the 222 cosets $c + \Sigma$. The smallest words of the 34 cosets of Σ in D that carry no new point are

7973, 34966, 132351, 133997, 136264, 138202, 164859, 171230, 527576, 529405, 562425, 564075, 657333, 657447, 661250, 692003, 692230, 1051583, 1082153, 1086366, 1088524, 1179858, 1187831, 1214532, 1218803, 1575157, 1578960, 1579074, 1605731, 1609713, 1707018, 1739534, 1741867, 1743801.

The magnitudes of the entries of $s_1, \ldots, s_{222}$ are in the accompanying data. Table 5 prints the first representative s_i for each pair of cosets of C swapped by w. Each is written as three rows of seven entries, coordinates 1 to 21. The representative s_1, with sign word 0, is the reweighted all-ones vector. The squared norms are $|s_1|^2 = 1\,000\,000\,001\,239\,123\,512$, $|s_3|^2 = 1\,000\,000\,000\,777\,524\,531$, $|s_5|^2 = 999\,999\,998\,470\,219\,855$ and $|s_7|^2 = 1\,000\,000\,001\,096\,310\,315$.

TABLE 5. One representative s_i for each pair of cosets of C in dimension 21.

s_1, cosets 0, 696	234431849	235593120	234431849	152856018	234431849	235305873	152344297
	233894149	233694134	235262955	153884776	235843572	152853654	237672411
	233891785	234651594	235395469	154135228	235173359	234521445	233694134
s_3, cosets 1393, 1993	−164500789	232437949	164500789	231876079	−157710920	−236489236	−235452527
	229320077	−229881947	238916196	−229881947	232437949	235452527	228283368
	235452527	235452527	162902210	157331629	232437949	233853948	232437949
s_5, cosets 2139, 2787	−153647918	−153647918	235341461	−238913097	−234664771	234664771	−233728926
	152852991	234523853	233799385	152035383	−234594312	234523853	233658467
	233728926	235271002	235411920	235271002	235341461	153647918	234594312
s_7, cosets 3370, 3986	234730343	−234478569	234713808	−234478569	153880863	−234478569	155028721
	235911272	−239871831	234478569	−234730343	−234495104	233330710	153959025
	234495104	153864328	234713808	233330710	233330710	153847793	233330710

REFERENCES

[1] griffinwork40, *D21-Kissing-31259*, GitHub repository, commit `aad7c5b`, 9 October 2026, `https://github.com/griffinwork40/D21-Kissing-31259`.

[2] H. Cohn, *Kissing numbers*, table, `https://cohn.mit.edu/kissing-numbers/`, accessed 4 October 2026.

[3] H. Cohn and A. Li, *Improved kissing numbers in seventeen through twenty-one dimensions*, preprint, 2024, version 3, 2026. arXiv:2411.04916v3.

[4] J. H. Conway and N. J. A. Sloane, *Sphere packings, lattices and groups*, third edition, Grundlehren der mathematischen Wissenschaften 290, Springer, New York, 1999.

[5] B. S. Ho, *A new lower bound for the kissing number in 19 dimensions*, preprint, 2026. arXiv:2603.10425v2.

[6] J. Leech, *Notes on sphere packings*, Canadian J. Math. **19** (1967), 251–267.

[7] J. Leech and N. J. A. Sloane, *Sphere packings and error-correcting codes*, Canadian J. Math. **23** (1971), no. 4, 718–745.

[8] N. Leijenhorst and D. de Laat, *Solving clustered low-rank semidefinite programs arising from polynomial optimization*, Math. Program. Comput. **16** (2024), no. 3, 503–534.

[9] F. J. MacWilliams and N. J. A. Sloane, *The theory of error-correcting codes*, North-Holland, Amsterdam, 1977.